

Linear Search

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Int arr[9]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
33	55	88	77	44	11	66	22	99

- 1) Get the **key** from the user.
- 2) Start the traversal from the 1st element(0th index).
- 3) Compare the key with every element of the array.
 - a. If the key is matching, return the corresponding index.
 - b. Else, continue the search till the last index.

```
for(i=0; i<9; i++)  
{  
    if(key == arr[i])  
        return i;  
}
```

Key = 11
Key found at index 5
Comparisons = 6

Average Case Time
Complexity
 $O(n/2)$ --> discard
 $1/2 \rightarrow O(n)$

Big Theta(n)
 $\Theta(n)$

Key = 33
Key found at index 0
Comparisons = 1

Best Case Time
Complexity
Order Of(1) --> $O(1)$

Big Omega(1)
 $\Omega(1)$

Key = 99
Key found at index 8
Comparisons = 9

Worst Case Time
Complexity
Order Of(n) --> $O(n)$

Big Oh(n)
 $O(n)$

Key = 7
Key Not Found
Comparisons = 9

Worst Case Time
Complexity
Order Of(n) --> $O(n)$

Big Oh(n)
 $O(n)$